

DR. EWE WIN ENG

Data Scientist | Machine Learning Engineer | Commercial AI Systems

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Visa Status: UK Global Talent Visa Holder (No sponsorship required) ILR in progress

PROFESSIONAL SUMMARY

Ph.D.-qualified Data Scientist and Quantitative Researcher with expert proficiency in Python, TensorFlow, and Time-Series Forecasting. Specialises in bridging theoretical physics and commercial AI applications, with a proven track record of optimising complex energy systems and supporting high-stakes public sector initiatives. Combining statistical rigour with production-level engineering, seeking to drive commercial value in Glasgow's Data Science or FinTech sectors. Recognised by the UK government as a leader in digital technology and science (UK Global Talent Visa Holder).

TECHNICAL SKILLS

- **Cloud Platforms:** Google Cloud Platform (GCP), Amazon Web Services (AWS)
 - **Databases & Data Processing:** PostgreSQL, PySpark, Pandas, Numpy
 - **DevOps & Tools:** Git, GitHub, Jupyter, Google Colab, Docker, FastAPI, Visual Studio Code
 - **Languages & Core:** Python, SQL, Bash Scripting.
 - **Machine Learning:** TensorFlow, Keras, Scikit-Learn, PyTorch, Deep Learning (ResNet-MLP), Genetic Algorithms, Random Forest, Decision Tree, XGBoost, Linear Regression, Softmax Regression, Logistic Regression, Neural Networks
 - **Data Visualisation:** Matplotlib, Seaborn, PowerBI
 - **Analytics & Maths:** Time-Series Forecasting, A/B Testing, Hypothesis Testing, Finite Volume Method, Regression & Classification Analysis, Optimization
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PROFESSIONAL EXPERIENCE

University of Strathclyde | Glasgow, UK | Postdoctoral Researcher | 2023 – 2025

- Act as an Energy Data Scientist and Modeller to engineer a commercial-grade 1D finite volume simulation engine in Python to model thermodynamic stratification, validating GigaWatt-hour thermal storage feasibility for industrial assets.
- Developed intelligent control algorithms for Heat Pump/CHP systems, linking grid balancing incentives directly to storage charging logic to maximise revenue capture.
- Conducted comprehensive techno-economic analysis (LCOH, COP) to de-risk the conversion of legacy mine shafts into renewable district heating assets.

- Built automated ETL pipelines using Pandas and SciPy to process high-resolution electricity demand profiles, enabling robust sensitivity analysis across varying tariff structures.

Universiti Kebangsaan Malaysia | Malaysia | Research Assistant | 2018 – 2022

- Optimized PVT systems by designing and implementing custom Genetic Algorithms to iterate through thousands of design variables, simulating performance outcomes for solar collectors.
 - Identified an optimal system configuration that increased energy capture efficiency by 30%, demonstrating direct ROI through algorithmic optimisation.
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KEY TECHNICAL PROJECTS

Automated Valuation Engine (Diamond Price Prediction) | Machine Learning Engineer

- Engineered a production-ready valuation pipeline utilizing a custom ResNet-MLP architecture to predict asset prices with high precision (R^2 more than 0.95).
- Implemented residual skip connections and Log-Norm target engineering to model complex non-linear interactions and stabilise gradients during training.

NHS Covid-19 Demand Modelling | Data Scientist

- Analyzed official NHS Scotland open data to model ICU bed usage
 - Developed a statistical framework for patient intake projection, demonstrating how data-driven insights support public health resource allocation.
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EDUCATION

- **Ph.D. in Renewable Energy** | Universiti Kebangsaan Malaysia | 2018 – 2022
 - **M.Sc. in Energy Technology** | Universiti Kebangsaan Malaysia | 2017 – 2018
 - **B.Sc. in Physics** | University of Glasgow, UK | 2014 – 2017
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CERTIFICATIONS

- **Build Data Lakes & Data Warehouses on Google Cloud** | Google | 2026
- **Generative AI Leader Specialization** | Google | 2026
- **Google AI Specialization** | Google | 2026
- **Python Project for Data Science** | IBM | 2025
- **Python for Data Science, AI & Development** | IBM | 2025
- **Data Visualization and Dashboards with Excel and Cognos** | IBM | 2025
- **Getting Started with Power BI Desktop** | Coursera | 2024